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Week 3 Simulator Breakpoint Specification

ORBIT NORTH · Week 3 · simulator scenario states

ORBIT NORTH

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Week 3 — Simulator Breakpoint Specification

This document specifies the four simulator states that ORBIT NORTH Week 3 needs. Three of them already exist as scenario YAML files in `~/SpaceMissionSimulation/configs/eosat1/scenarios/`. One needs to be created.

For each breakpoint we list: - **Purpose** — which Week 3 session it serves - **Mission phase** — EOSAT-1 phase number (0=PRE_SEPARATION, 3=BOOTLOADER_OPS, 4=LEOP, 5=COMMISSIONING, 6=NOMINAL) - **Spacecraft state** — power lines, AOCS mode, antennas, payload, battery - **Pass geometry** — which ground station, AOS minus how many minutes - **Existing scenario** — YAML file path to use, or **NEW**

BP-1 — Nominal Post-Commissioning *(existing)*

Purpose. Training Day 1 round-robin, Training Day 2 dry-runs, and the *nominal-shift* example used for procedure familiarisation.

Mission phase. 6 (NOMINAL).

Spacecraft state.

- All antennas deployed; TT&C on the LOW rung (16 kbps) for routine contacts, HIGH (128 kbps) only for dumps above 10° elevation.
- AOCS mode: NOMINAL (4, nadir pointing), body rates < 0.01 °/s, reaction wheels spun up.
- EPS: battery 90% SoC, solar arrays nominal, all main power lines ON.
- TCS: heaters cycling on thermostat, all zones in green band.
- Payload: imager in STANDBY, FPA cooler OFF (STANDBY does not run it; NOM-002 switches it on about 10 min before an imaging window).
- OBDH: application image running (`obdh.sw_image` = 1), FDIR enabled.

Pass geometry. The default in `nominal_ops.yaml` lets the orbit propagator generate natural passes. For Training Day 1 the instructor selects a starting epoch ~10 min before the next Iqaluit AOS so students see a full pass start to LOS.

Existing scenario. `scenarios/nominal_ops.yaml`. No change required.

BP-2 — LEOP Day Start *(existing)*

Purpose. Wednesday 19 August simulation day.

Mission phase. 3 (BOOTLOADER_OPS).

Spacecraft state.

- OBC running bootloader firmware, transmitting beacon TM (SID 11 only) on the bootloader APID.
- All other power lines OFF: ttc_tx is in beacon-only mode, aocs_wheels OFF, payload OFF, fpa_cooler OFF, htr_bat OFF, htr_obc OFF.
- Antennas NOT yet deployed — only the patch antenna for beacon is available.
- AOCS in OFF mode with residual tumble rates ~1–2 °/s from the separation impulse (already partially decayed).

- EPS battery at ~85% SoC, solar arrays nominal but uncontrolled Sun-pointing.
- TCS: passive only, all zones at ~ambient.

Pass geometry. Iqaluit pass beginning at scenario time T+0; the instructor sets the simulator clock such that AOS occurs at scenario start.

Existing scenario. `scenarios/first_contact.yaml`. The `override_passes: true` flag in that file ensures continuous visibility for the duration the team needs.

Operator pass criteria for the day. By end of the second pass (team swap), the spacecraft must be in phase 4 (LEOP), antennas deployed, app SW booted, time set, and all power lines required for detumble (htr_bat, aocs_wheels) ON.

BP-3 — Commissioning Day Start *(NEW — must be created)*

Purpose. Thursday 20 August simulation day. The mission has just exited LEOP — antennas are deployed, app SW is running, detumble has completed, time is set, but no commissioning has begun.

Mission phase. 4 (LEOP-complete) → transitions to 5 (COMMISSIONING) once team commands the phase change.

Spacecraft state required at scenario start.

- All antennas DEPLOYED.
- App SW RUNNING, full HK active (all SIDs 1–N reporting at 4 s rate).
- AOCS mode: DETUMBLE-completed, body rates < 0.05 °/s, reaction wheels POWERED but at 0 RPM.
- EPS battery 90% SoC; main bus nominal; payload OFF; fpa_cooler OFF; htr_bat ON; htr_obc ON; htr_payload OFF.
- TCS: payload zone at ~+5 °C (no active control yet); FPA at ambient.
- OBDH: phase 4, FDIR partially configured (LEOP-only set), nominal FDIR not yet enabled.
- Time: onboard clock synced.

Pass geometry. First Iqaluit pass beginning at scenario T+0.

Suggested YAML to add to the simulator — `configs/eosat1/scenarios/commissioning_day_start.yaml`:

```
name: "Commissioning Day Start"
difficulty: INTERMEDIATE
duration_s: 14400          # 4 hours of compressed time across multiple passes
briefing: |
  EOSAT-1 has just exited LEOP. Antennas are deployed, app SW is
  running, detumble has completed, time is set, and the spacecraft
  is in phase 4 with FDIR configured for the LEOP-only fault set.
  Your shift covers the first 4 hours of commissioning across
  ~4 ground passes (compressed).
  Goals for the shift:    1. Transition the spacecraft to phase 5 (COMMISSIONING) if LEOP-
    3. Power on the payload, cool the FPA, and complete COM-012
       (First Light) before LOS at end of pass 4.

initial_conditions:
  spacecraft_phase: 4
```

```
  override_passes: true
  soc_pct: 90.0
  power_lines:
    ttc_tx: true
    htr_bat: true
    htr_obc: true
    aocs_wheels: true
    payload: false
    fpa_cooler: false
    prop: false
  params:    0x020F: 3        # aocs.mode = COARSE_SUN (COM-004 enables the wheels and com
    0x0600: 0        # payload.mode = OFF

events:
  - time_offset_s: 600
    action: message
    params:
      text: "Optional: easy anomaly - inject eps overcurrent on line 3 during COM-009 payl

  - time_offset_s: 5400
    action: message
    params:
      text: "Optional: medium anomaly - inject tcs cooler_failure during COM-010 FPA cool-

expected_responses:
  - { category: command, description: "Declare COMMISSIONING: SYS_SET_MISSION_PHASE(phase=
  - { category: command, description: "Run COM-001 EPS Checkout" }  - { category: command,
  - { category: command, description: "Run COM-005 AOCS Mode Transitions" }
  - { category: command, description: "Run COM-006 TTC Link Verification" }
  - { category: command, description: "Run COM-008 FDIR Configuration — enable S12 monitor
  - { category: command, description: "Run COM-009 Payload Power On" }
  - { category: command, description: "Run COM-010 FPA Cooler Activation" }
  - { category: command, description: "Run COM-012 First Light — capture and assess one im
```

Operator pass criteria for the day. Two of the four passes must include FPA cooler ON; First Light must be acquired or formally deferred-with-justification by end of pass 4.

BP-4 — End-of-Sim "What If" Branch *(optional)*

Purpose. Friday wrap. Lets the instructor replay a Wednesday or Thursday decision-point with a different team choice, to make the debrief concrete.

Implementation. The simulator already supports state snapshots. Instructor takes a snapshot at the moment of the decision (e.g. just before the team commits to safe-mode entry); during Friday wrap, the sim is restored from that snapshot and a different team commands the alternative branch. No new scenario file required.

Procedure. There is no `sim` command-line tool. Breakpoints are saved and restored from the instructor console (the **Breakpoints** panel: SAVE, then LOAD against the saved name) or over the instructor HTTP API on the simulator port (default 8080, `$SMO_SIM_HTTP_PORT`):

```
curl -X POST http://localhost:8080/api/breakpoint/save \
     -H 'Content-Type: application/json' -d '{"name": "snap_<scenario>_<event>"}'
curl -X POST http://localhost:8080/api/breakpoint/load \
     -H 'Content-Type: application/json' -d '{"name": "snap_<scenario>_<event>"}'
curl http://localhost:8080/api/breakpoints
```

A saved breakpoint is written to `workspace/breakpoints/<name>.json` under the simulator's working directory, so it survives a restart. Save and load both run between simulation ticks, so a restore never lands mid-tick. See the instructor book, §5.10.1.

Appendix — Scenario file conventions

EOSAT-1 scenario YAML structure (consistent across all files in `scenarios/`):

Key	Meaning
`name`	Display name
`difficulty`	BASIC / INTERMEDIATE / ADVANCED
`duration_s`	Sim seconds (compressed time multiplier defined separately)
`briefing`	Multiline operator briefing text
`initial_conditions`	Spacecraft phase, optional state overrides, optional speed multiplier
`events`	List of timed instructor or auto events
`expected_responses`	Operator action checklist used by the assessor

The new `commissioning_day_start.yaml` follows this convention.